



A novel data augmentation approach to fault diagnosis with class-imbalance problem

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ARTICLE INFO

Keywords:

Class imbalance
Fault diagnosis
Conditional variational auto-encoder
Kullback–Leibler Divergence Vanishing
Kernel mean matching

ABSTRACT

Data-driven fault diagnosis techniques are frequently applied to ensure the reliability and safety of industrial systems. However, as a common challenge, the class-imbalance problem reduces the performance of data-driven methods due to the lack of data information. We propose a weighted modified conditional variational auto-encoder (WM-CVAE) as a novel data augmentation technique to tackle the issue. The modified structure can alleviate the existing Kullback–Leibler (KL) divergence vanishing by an adaptive loss. Meanwhile, kernel mean matching (KMM) is proposed on weight computation to reduce the negative effect of dissimilar generated samples. Constructing the WM-CVAE data augmentation framework can effectively improve the data quality and learning capability in class-imbalance fault diagnosis. To validate the proposed WM-CVAE model, three real-world industrial datasets are used as study objects, and the random forest is used as the base learner in the fault classification tasks. The diagnostic results demonstrate that the proposed WM-CVAE data augmentation framework can improve learning results in class-imbalance fault diagnosis.

1. Introduction

Fault diagnosis is a crucial technique to ensure the reliability and safety of complex industrial systems and enables real-time monitoring through process sensor data [1–3]. High-performance fault diagnosis is essential for condition-based maintenance, including model-based and data-driven approaches [4]. Traditional model-based methods rely on physical mechanisms, which are challenging to provide reliable information under complex conditions [5]. Data-driven fault diagnosis methods rely on intelligent algorithms and historical data from sensors [6,7]. Due to complex mechanisms and advanced computer science, data-driven approaches are welcome in academia and industry [8,9]. The success of deep learning in various fields has also provided impetus in this direction [10,11].

However, data-driven methods are less accurate in dealing with imperfect datasets, which limits their applications in real-world industrial systems [12,13]. Class-imbalance is one of the hot problems due to the limited sampling environment and conditions in the actual data acquisition process [14]. The exact reason for this problem is that machines usually work typically and have fewer faults. Meanwhile, common faults occur more frequently than rare ones due to different mechanisms of occurrence [15]. From the perspective of data-driven methods, the class-imbalance problem biases classification decisions toward the majority class [16]. It will lead to a decrease in the accuracy

of the minority samples (fault categories) that we pay more attention to, thereby affecting the actual effectiveness of the model.

Existing improvement methods for class-imbalance problems are divided into four directions: data resampling, feature representation, algorithm improvement, and ensemble learning [17]. Data resampling approaches rely on generative model-based data over-sampling and majority sample reduction-based under-sampling, etc. Classic over-sampling strategies, such as the Synthetic Minority Oversampling Technique (SMOTE) and its variants, have been used widely in class-imbalance problems [18]. Some novel over-sampling methods, including generative adversarial networks (GAN) [19] and variational auto-encoder (VAE) [20,21], can often achieve better results by complex nonlinear capabilities compared with classic methods [22]. Under-sampling methods reduce the majority of samples with the help of metric information [23]. Qian et al. proposed an adaptive under-sampling method to tackle the class-imbalance fault detection problem [17]. However, under-sampling methods often perform poorly under rare minority samples. From the feature representation point of view, the learner can extract a new group of features, which is more favorable for class-imbalance model training and enhances diagnostic performance. Li et al. used a joint attention feature transfer network for gearbox class-imbalance fault diagnosis [24]. Some viewpoints hold that the generalization and robustness of the learners can be improved by

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<https://doi.org/10.1016/j.ress.2023.109832>

Received 29 June 2023; Received in revised form 14 November 2023; Accepted 16 November 2023

Available online 19 November 2023

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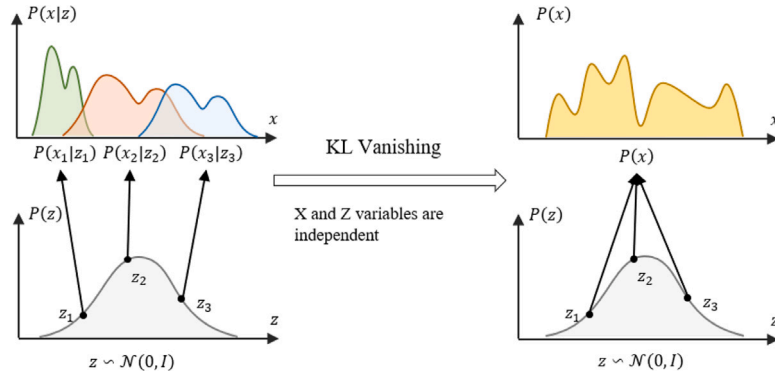


Fig. 1. KL divergence vanishing problem of VAE/CVAE method.

some optimized methods, resulting in better performance under class-imbalance problems. A higher penalty weight on the minority class of loss function can also make the learner pay more attention to minority classes. Khan et al. proposed a cost-sensitive convolutional neural network [25]. Wu et al. designed a cost-sensitive deep domain adaptation network under variant working and class-imbalance conditions [26]. Some ensemble learning algorithms have been applied to tackle class-imbalance problems. Qian et al. proposed a boosting-based ensemble framework (BEF) with a weakly supervised technique based on the cost-sensitive neighborhood component analysis [27]. It can reduce the overfitting negative impact of the traditional methods and achieve better results.

The lack of minority class data is a fundamental problem for class-imbalance fault diagnosis, so the data resampling approaches are the hot direction. Conventional data over-sampling/under-sampling based methods cannot handle complex high-dimensional data augmentation. Based on deep learning-based techniques, the VAE-based methods can leverage the powerful nonlinear fitting capability to compensate for the shortcomings of traditional methods and achieve better data augmentation [28]. An encoder can output approximately normally distributed latent variables based on raw data samples [29]. Meanwhile, the data samples are reconstructed employing the decoder. Under this exquisite design, the VAE-based method can better extract the internal information of the data and then generate similar data. As a variant of VAE, CVAE is proposed in this paper to realize data augmentation conveniently. There are already many VAE/CVAE-based studies to expand data information. Khanam et al. proposed a VAE-based detection model by generating new samples and got effective results [30]. Zhao et al. proposed a normalized CVAE-based method with adaptive focal loss towards the bearing-rotor systems class-imbalance fault diagnosis [31]. Therefore, VAE/CVAE-based models can effectively improve the performance of fault diagnosis and achieve the reliability and safety of industrial systems. However, such VAE and CVAE methods suffer from two critical issues in data generation, including the KL divergence vanishing and low similarity of generated samples [32], which are our main technical motivations. The detailed illustrations are as follows.

(1) **KL divergence vanishing:** Due to the trained strong decoders of VAE/CVAE methods, a loss term of KL divergence will gradually vanish with the training process until it approaches 0, which is called KL vanishing (some paper named as ‘posterior collapse’) [33]. It causes a weak link between the original and generated data, resulting in low-quality samples with large reconstruction errors. The diagram of KL divergence vanishing is shown in Fig. 1. Few papers focus on the KL divergence vanishing, and some solutions include instance-based methods (Batch Normalization) [34], δ -loss function [35], etc. These methods focus more on the reconstructing error term during model training to avoid the KL vanishing. In this paper, inspired by the above ideas, we propose a modified CVAE method based on an adaptive weight loss function to solve the problem.

(2) **Insufficient similarity of generated samples:** Noise is widely present in industrial systems, which can negatively affect the quality of samples generated by the CVAE-based data augmentation approach. To determine the quality of the samples produced, we apply the KMM algorithm to calculate the weights of the generated data [36]. Higher weights are given to the more similar data, reducing the effect of dissimilar data during training.

In this paper, based on the two main motivations, we propose a novel CVAE-based data augmentation to relieve these harmful effects. To test the validity of the data augmentation technique, we utilize the random forest classifier as the base learner due to its high accuracy and strong generalization ability. The main contributions of this model can be summarized as follows.

- We propose a general adaptive loss-based learning algorithm to solve the KL divergence vanishing problem widely existing in the original CVAE generators.
- An adaptive weight case is proposed as an example to be analyzed and validated on real-world class-imbalance datasets, including the loss function curves, t-SNE visualization, and experimental results.
- As a weight assignment algorithm, the KMM method is applied to assign larger weights on similar generated samples and reduce the negative effects of dissimilar samples.
- The experimental results in three real-world datasets indicate the effectiveness of the proposed method under class-imbalance scenarios, which can ensure the reliability and safety of industrial systems.

The remainder of this paper is organized as follows. Section 2 reviews related works. Section 3 describes the proposed WM-CVAE data augmentation method. Section 4 presents results and analysis of real-world case studies. Finally, Section 5 concludes the paper.

2. Related works on CVAE method

Auto-encoder (AE) can reconstruct samples by model training and has a wide application. A hidden layer inside AE model can produce code z with the help of encoder $z = f(x)$ and decoder $x' = g(z)$. VAE is an improved version of AE [37], and the encoder can compress the input data x to a latent variable z with a certain distribution. A recognition model $q_\phi(z|x)$ with variation parameter ϕ is used to estimate the approximate complex true prior distribution $p_\theta(z|x)$ with generation parameter θ . During the training process, the encoder-generated z under input data x is a latent variable representation, with a certain distribution. In VAE models, set the latent variable to follow normal Gaussian distribution $z \sim \mathcal{N}(0, I)$. For a probabilistic decoder, given a set of latent variables z , it can generate a large number of new data points x' , and can still generate the corresponding distribution $p_\theta(x|z)$.

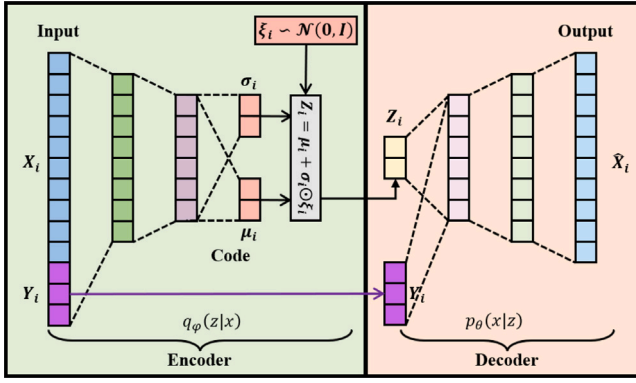


Fig. 2. Detailed structure of conditional variational auto-encoder method.

The primary purpose of VAE is to enlarge the marginal likelihood term, as shown in Eq. (1).

$$\sum_{i=1}^N \log p_{\theta}(x_i) = \sum_{i=1}^N KL(q_{\phi}(z|x_i) \parallel p_{\theta}(z|x_i)) + ELBO(\theta, \phi, x_i) \quad (1)$$

The first term of the RHS (right-hand side) uses the KL divergence to represent the difference between the approximate and the actual model distribution. The second RHS term is the Evidence Lower Bound (ELBO), which represents the lower bound of the maximum likelihood term at data point i . The purpose of VAE/CVAE can be transferred to maximize the ELBO value as shown in Eq. (2):

$$\begin{aligned} \theta_*, \phi_* &\approx \underset{\theta, \phi}{\operatorname{argmax}} \left(\sum_i ELBO(\theta, \phi, x_i) \right) \\ &\approx \underset{\theta, \phi}{\operatorname{argmax}} \left(- \sum_{i=1}^N KL(q_{\phi}(z|x_i) \parallel p_{\theta}(z|x_i)) + \sum_{i=1}^N \log p_{\theta}(x_i) \right) \\ &\approx \underset{\theta, \phi}{\operatorname{argmax}} \left(- \sum_{i=1}^N KL(q_{\phi}(z|x_i) \parallel p_{\theta}(z)) + \sum_{i=1}^N [\log p_{\theta}(x_i|z_i^l)] \right) \end{aligned} \quad (2)$$

where z_i^l can be sampled by generated model $g_{\phi}(x_i)$ with L samplings. In VAE/CVAE model, there is an assumption of prior distribution $p_{\theta} = \mathcal{N}(0, I)$, which is a multivariate normal Gaussian distribution. And there is a learned approximate recognition model $q_{\phi}(z|x_i) = \mathcal{N}(\mu_i, (\sigma_i)^2 I)$. Then, z_i is sampled by points of Gaussian distribution with mean μ_i and variance σ_i , using equation $z_i^l = \mu_i + \sigma_i \odot \xi_i$, where $\xi_i \sim \mathcal{N}(0, I)$, and $\odot$ represents element-wise dot production.

CVAE is an extended version of VAE methods with a known condition label y [38], and the detailed structure is shown in Fig. 2. The CVAE comprises an encoder and decoder network by maximizing the ELBO term, which can be extended as Eq. (3).

$$ELBO(\theta, \phi, x_i, y_i) = -KL(q_{\phi}(z|x_i, y_i) \parallel p_{\theta}(z)) + \frac{1}{N} \sum_{l=1}^N [\log p_{\theta}(x_i|z_i^l, y_i)] \quad (3)$$

3. Methodology

In this section, we introduce a novel CVAE-based data augmentation method, including the adaptive loss and weight calculation of generated samples.

3.1. Modified CVAE model

For CVAE neural networks, the loss function can be set as Eq. (4). Then, the maximum of ELBO term can be equivalent to the minimum of $L_{\theta, \phi}$.

$$L_{\theta, \phi} = -ELBO(\theta, \phi, x_i, y_i) = L_{Recon} + L_{\mu, \sigma^2} \quad (4)$$

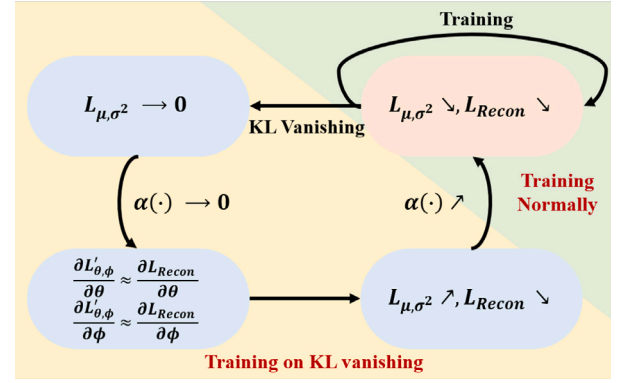


Fig. 3. Effect of adaptive weights during training when KL vanishing.

where L_{Recon} is the reconstruction error of input and output, L_{μ, σ^2} means the KL divergence penalty term. The detailed expressions can be shown in Eqs. (5) to (6).

$$L_{Recon} = -\frac{1}{N} \sum_{i=1}^N \log(P_{\theta}(x_i|z_i, y_i)) \quad (5)$$

$$\begin{aligned} L_{\mu, \sigma^2} &= KL(q_{\phi}(z|x_i, y_i) \parallel p_{\theta}(z)) = KL(\mathcal{N}(\mu, \sigma^2) \parallel \mathcal{N}(0, I)) \\ &= \int \mathcal{N}(\mu, \sigma^2) \log \frac{\mathcal{N}(\mu, \sigma^2)}{\mathcal{N}(0, I)} dx = \frac{1}{2} \sum_{i=1}^N (-\log(\sigma_i^2) + \mu_i^2 + \sigma_i^2 - 1) \end{aligned} \quad (6)$$

Although the CVAE model has two optimization objectives, it often insists on minimizing the KL divergence term close to zero in actual execution. Compared with the reconstruction task, this task is simpler for the CVAE model. Excessive pursuit of minimizing one task can lead to a performance drop for another task, thereby affecting the quality of generated samples. The decoder ignores the latent space of z and reconstruct new data points $\hat{x} \sim P_{\theta}(x_i|y)$ instead of $P_{\theta}(x_i|y, z)$. Then, KL divergence term L_{μ, σ^2} will vanish to zero.

Based on the problem, this paper propose an adaptive weight α -based CVAE model, which can add constraints to the KL penalty term L_{μ, σ^2} , as shown in Eq. (7):

$$L'_{\theta, \phi} = L_{Recon} + \alpha(L_{Recon}, L_{\mu, \sigma^2}) L_{\mu, \sigma^2} \quad (7)$$

where $\alpha(L_{Recon}, L_{\mu, \sigma^2})$ is an adaptive loss weight for the CVAE model, and its design ideas can be shown in Fig. 3. Like a adaptive switch, it controls the weight of KL penalty term during the training process. Based on the above idea, a special adaptive weight case is proposed here as an example to be analyzed, and its detailed form is as shown in Eq. (8).

$$\alpha(L_{Recon}, L_{\mu, \sigma^2}) = \frac{L_{\mu, \sigma^2}}{L_{Recon} + L_{\mu, \sigma^2}} \quad (8)$$

It can let models more inclined to reconstruction loss term L_{Recon} when the KL divergence is close to zero, which can avoid the KL vanishing issues caused by KL penalty terms approaching 0. In the CVAE approach, ϕ and θ are the model parameters of the encoder and decoder. Their gradient descent patterns are shown in Eqs. (9) to (10):

$$\phi^{(t+1)} = \phi^{(t)} - r \frac{\partial L'_{\theta, \phi}}{\partial \phi} \quad (9)$$

$$\theta^{(t+1)} = \theta^{(t)} - r \frac{\partial L'_{\theta, \phi}}{\partial \theta} \quad (10)$$

where $\phi^{(t+1)}$ and $\theta^{(t+1)}$ mean the $(t+1)$ th epoch of parameters ϕ and θ . r represents the learning rate. When we use the adaptive loss as shown in Eq. (7), the gradient descent term can be extended by chain rule,

and the detailed formats are shown in Eqs. (11) to (12).

$$\frac{\partial L'_{\theta,\phi}}{\partial \phi} = \frac{\partial L_{Recon}}{\partial \phi} + \underbrace{\alpha(\cdot) \frac{\partial L_{\mu,\sigma^2}}{\partial \phi} + L_{\mu,\sigma^2} \frac{\partial \alpha(\cdot)}{\partial \phi}}_{\Phi(L_{\mu,\sigma^2})} \quad (11)$$

$$\frac{\partial L'_{\theta,\phi}}{\partial \theta} = \frac{\partial L_{Recon}}{\partial \theta} + \underbrace{\alpha(\cdot) \frac{\partial L_{\mu,\sigma^2}}{\partial \theta} + L_{\mu,\sigma^2} \frac{\partial \alpha(\cdot)}{\partial \theta}}_{\Theta(L_{\mu,\sigma^2})} \quad (12)$$

When $L_{\mu,\sigma^2} \rightarrow 0$, the $\Phi(L_{\mu,\sigma^2})$ and $\Theta(L_{\mu,\sigma^2})$ terms are close to 0 due to their special structure. Therefore, Eqs. (11) to (12) can be simplified as only one term related to reconstruction error L_{Recon} . In this case, the CVAE model degenerates into an AE model, and the gradient descent formulas can be approximated as Eqs. (13) to (14).

$$\phi^{(t+1)} \approx \phi^{(t)} - r \frac{\partial L_{Recon}}{\partial \phi} \quad (13)$$

$$\theta^{(t+1)} \approx \theta^{(t)} - r \frac{\partial L_{Recon}}{\partial \theta} \quad (14)$$

The model focuses on reducing reconstruction errors, and the KL divergence term increases by no constraint. Therefore, the adaptive loss can avoid the KL vanishing problem to enhance the generated samples quality. When $L_{\mu,\sigma^2} \neq 0$, the gradient descent formulas return to the normal condition, and the approximate AE model also changes to CVAE model.

3.2. Weight calculation of generated samples

Inspired by metric learning [39], we conduct a KMM-based weight calculation work to take the distribution discrepancy between generated and real samples as knowledge, and the motivation is to reduce the discrepancy. Therefore, it can reduce the negative effect of poor similarity between the generated data X_G and the original data X_O .

Define the mapping $\psi(x)$ from data points x to a reproducing kernel Hilbert space (RKHS). A feature representation of distribution can be expressed as $E_{x \sim P(x)}[\psi(x)]$, represents the expectation of samples with $P(x)$ distribution in RKHS. Each generated sample has a weight ratio β . To find the value of β , KMM is used to minimize the Eq. (15).

$$\beta^* = \arg \min_{\beta} \|E_{x \sim P_G(x)}[\beta_i \psi(x)] - E_{x \sim P_O(x)}[\psi(x)]\|_H^2 \quad (15)$$

s.t. $\beta \geq 0, |E_{x \sim P_G(x)}[\beta_i] - 1| \leq \epsilon$

It can be extended as the Eq. (16):

$$\begin{aligned} \beta^* &= \arg \min_{\beta} \left\| \frac{1}{N_G} \sum_{i=1}^{N_G} \beta_i \psi(x_i) - \frac{1}{N_O} \sum_{i=1}^{N_O} \psi(x'_i) \right\|_H^2 \\ &= \arg \min_{\beta} \left(\sum_{i=1}^{N_G} \sum_{j=1}^{N_G} \frac{\beta_i \beta_j}{N_G^2} \kappa(x_i, x_j) - 2 \sum_{i=1}^{N_G} \sum_{j=1}^{N_O} \frac{\beta_i}{N_G N_O} \kappa(x_i, x'_j) \right. \\ &\quad \left. + \sum_{i=1}^{N_O} \sum_{j=1}^{N_O} \frac{1}{N_O^2} \kappa(x'_i, x'_j) \right) \\ &= \arg \min_{\beta} \left(\frac{1}{N_G^2} \beta^T K \beta - \frac{2}{N_G N_O} K_2^T \beta + \text{const} \right) \\ &= \arg \min_{\beta} \left(\frac{1}{N_G^2} \beta^T K \beta - \frac{2}{N_G^2} K_*^T \beta \right) \end{aligned} \quad (16)$$

where N_G and N_O are samples number of generated and original domain, $\kappa(x_i, x_j)$ means the kernel function, $\beta = [\beta_1, \dots, \beta_{N_G}]^T$, K with $\{K|K_{i,j} = \kappa(x_i, x_j)\}$, K_2 with $\{K_2|K_{2,(i,j)} = \sum_{i=1}^{N_O} \kappa(x_i, x'_j)\}$, and K_* with $\{K_*|K_{*,(i,j)} = \frac{N_G}{N_O} \sum_{i=1}^{N_O} \kappa(x_i, x'_j)\}$.

The weight calculation can be converted to a quadratic program, and solved by optimization procedure [36]. β is normalized the range from 0 to 1.

3.3. WM-CVAE-based fault diagnosis approach

The framework of the WM-CVAE data augmentation approach is shown in Fig. 4. It includes five main components, including training data input, encoder and decoder of CVAE generator, weight calculation, base learner training, and online classification task. We obtain the imbalanced historical training data from real-world sensors. The proposed generator is introduced in Sections 2 and 3.1, and the weight calculation is presented in Section 3.2. We conduct the base learner training and realize the classification tasks using the generated and real samples. The learning objective of the base fault classifier can be redefined as shown in Eq. (17):

Algorithm 1: Process of proposed WM-CVAE data augmentation approach

Input: Training datasets $\{X_{Tr,c_i}, Y_{Tr,c_i}\}$ with C category and N_i samples for each i th class. Test datasets $\{X_{Te}, Y_{Te}\}$

Output: Detailed model parameters of WM-CVAE-RF model, predicted results $\hat{Y}_{Test}$

- 1 Feature extraction and preprocessing;
- 2 Initialization and hyper-parameters settings;
- // Offline WM-CVAE data augmentation
- 3 **while** $j \leq Epoch_{max}$ **do**
- 4 Calculate $q_{\phi}(z|x_i) = \mathcal{N}(\mu_i, (\sigma_i)_2 I)$ by equation $z_i^l = \mu_i + \sigma_i \odot \xi_i$;
- 5 Calculate reconstructed data x'_i and adaptive loss function $L'_{\theta,\phi}$ by Eqs. (4) to (8);
- 6 Update the parameters ϕ and θ of the CVAE model using gradient descent by Eqs. (9) to (10);
- 7 $j \leftarrow j + 1$;
- 8 **end**
- 9 Get all generated samples $\{X_{G,c_i}, Y_{G,c_i}\}$ with $N_{max} + 1 - N_i$ numbers for faulty labels 1, 2, ..., C on $\{X_{Tr,c_i}, Y_{Tr,c_i}\}$;
- 10 Get new datasets $\{X_{new}, Y_{new}\}$ including generated and original data;
- 11 Calculate the weights $\hat{\beta}_i$ with every generated data by Eqs. (15) to (16);
- // Offline model training
- 12 Train the base classifier on new datasets with weights $\hat{\beta}_i$ by Eqs. (17) to (18);

$$\theta^* = \arg \min_{\theta} \left(\sum_i^{N_G} \hat{\beta}_i l(f_{x_i}, y_i) + \sum_i^{N_O} l(f_{x'_i}, y_i) \right) \quad (17)$$

where $l(f_{x_i}, y_i)$ is defined as the loss function. N_G and N_O represent the number of generated and original data. θ^* means the parameters of base classifier.

Random forest consists of p decision trees with a bootstrap sampling approach. After WM-CVAE data augmentation, datasets $\{X_{tr}, \beta, Y_{tr}\}$ can be used to train the p trees. During the training process, the trained $Tree_k$ can be obtained by Eq. (17). The classified results $Y_{te,k}$ can be got on the test data $X_{te} = \{x_1, \dots, x_{N_{test}}\}$. The majority classified label is set as the final classification result $RF(X_{te})$ during the *Vote* process, as shown in Eq. (18).

$$RF(X_{te}) = \underset{k=1,2,\dots,p}{Vote}(Y_{te,k}) = \underset{k=1,2,\dots,p}{Vote}(Tree_k(X_{te})) \quad (18)$$

The necessary steps of proposed WM-CVAE data augmentation approach are shown in Algorithm 1. Steps (1 to 2) are the necessary works for data processing and analysis. The data augmentation of modified CVAE method are shown in steps (3 to 8). Then, in steps (9 to 10), CVAE-based generator can get new augmented data and combine them with original data as a new dataset. To enhance the quality of generated

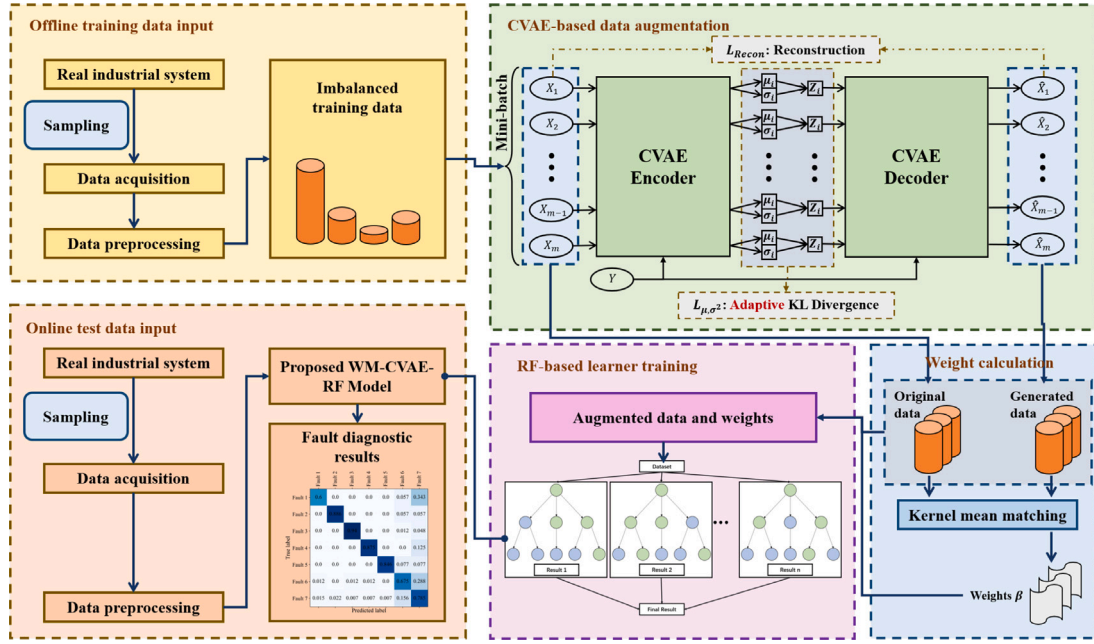


Fig. 4. Framework of proposed WM-CVAE-RF approach on class-imbalance fault diagnosis.

data points, KMM can calculate the weights in step (11). After the WM-CVAE data augmentation, random forest fault classifier is used to complete the classification task in step (12).

4. Case study of class-imbalance fault diagnosis

To verify the effectiveness of the proposed WM-CVAE data augmentation model, three industrial fault datasets are used for evaluation, including a steel plate dataset [40], a multi-phase flow process dataset [41], and a bearing dataset [42].

4.1. Performance evaluation settings

The imbalance rate Im_r is a crucial indicator to describe the imbalanced state of the classification task, as shown in Eq. (19):

$$Im_r = \frac{\max(N_c)}{\min(N_c)}, c = 1, 2, \dots \quad (19)$$

where N_c means the samples number of c class.

Three evaluation indicators are used in classification task, including recall, precision, and F1-score. Therefore, we use M_{Pr} , M_{Re} , and M_{F1} as the metric evaluator, and M_{F1} is the critical performance index among all terms.

4.2. Compared methods

We compare our proposed approach with several methods for solving the class-imbalance issues, and their introductions are shown below.

- ROSE [43]: Random over-sampling example.
- RUSE [43]: Random under-sampling example.
- SMOTE [22]: Synthetic Minority Over-Sampling Technique.
- ADASYN [44]: ADaptive SYNthetic Sampling Approach.
- MSDA [42]: A Mixed Signal Data Augmentation method with the help of noise.
- EM-LMOT [45]: Expected Maximization (EM)-based Local-weighted Minority Oversampling Technique (LMOT).
- EF-NCA [27]: A boosting-based Ensemble Framework (EF) with weakly supervised oversampling technique under a cost-sensitive Neighborhood Component Analysis (NCA).

Table 1

Description of training set for steel plate datasets with class-imbalance problems.

Class	No. of samples						Description
	Task 1	Task 2	Task 3	Task 4	Task 5	Task 6	
1	100	100	100	100	100	100	Pastry
2	100	100	100	100	100	100	Z-Scratch
3	100	100	100	100	100	100	K-Scratch
4	30	15	10	6	5	3	Stains
5	30	15	10	6	5	3	Dirtiness
6	300	300	300	300	300	300	Bumps
7	300	300	300	300	300	300	Other faults
Im_r	10	20	30	50	60	100	

- GAN [46]: Generative Adversarial Networks on data augmentation.
- CVAE [31]: Only use CVAE model on data augmentation.
- β -CVAE [47]: Use β for weighted CVAE model on data augmentation.

ROSE, RUSE, SMOTE, ADASYN, and MSDA are classical methods for solving imbalance problems, and EM-LMOT and EF-NCA are recent state-of-the-art methods. GAN and CVAE are neural networks-based generation methods. And β -CVAE is an effective method to solve the KL divergence vanishing.

During the implementation of these data-driven methods, data pre-processing is necessary to improve data quality. The missing values are filled with 0 values, and a min-max normalization is used to reduce the impact of absolute data size. To ensure the results are statistically significant, we conduct ten repeated experiments and take the average value to represent the classification results.

4.3. Case 1: Experiments on steel plates fault datasets

4.3.1. Datasets description and experimental settings

The steel plate faults data is supplied by the Research Center of Sciences of Communication and obtained in the UCL machine learning repository [40]. Steel plates are widely applied in the field of industrial manufacturing, and the related fault diagnosis work has become important. In the real-world dataset, 27 independent variables are collected

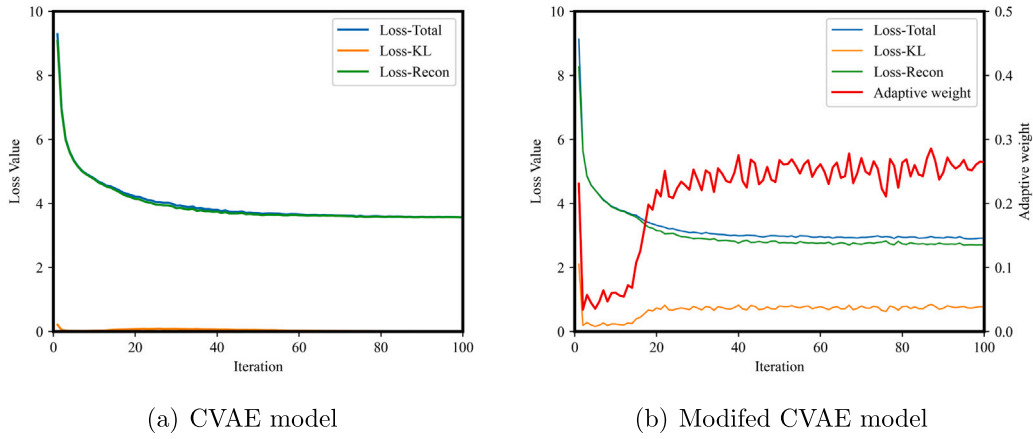


Fig. 5. Comparison of learning loss curves during training process.

Table 2

The adjustment results of network hyper-parameters.

Num	Nodes	Batch size	Learning rate	F1-score
1	128, 32, 8	128	0.001	0.7765
2	128, 32, 8	128	0.002	0.7859
3	128, 32, 8	128	0.005	0.7975
4	128, 32, 8	128	0.01	0.7724
5	128, 32, 8	128	0.02	0.7606
6	128, 32, 8	256	0.005	0.7913
7	128, 32, 8	64	0.005	0.7878
8	128, 32, 8	32	0.005	0.7797
9	256, 64, 16	128	0.005	0.7861
10	256, 64, 8	128	0.005	0.7837
11	128, 32, 4	128	0.005	0.7897
12	64, 32, 8	128	0.005	0.7873
13	32, 16, 8	128	0.005	0.7867
14	32, 16, 4	128	0.005	0.7916

by sensors of steel plates, and there are seven types of dependent faults: more samples for faults 6 and 7 and few samples for faults 4 and 5. We set up six fault classification tasks that correspond to different imbalance rates. Specific information about the fault description and training set numbers of six tasks can be shown in Table 1. On the test set, we uniformly set each faulty category to 25 samples without data augmentation methods to verify the algorithm's performance.

4.3.2. Adjusted hyper-parameter experiments and settings

Hyper-parameter settings are essential for neural networks-based algorithms, and the learning rate, batch size, and number of nodes are the critical hyper-parameters that need to be adjusted by experiments. To get better hyper-parameters, we do the adjustment experiment on Task 1, and the results are shown in Table 2. During the training process of WM-CVAE, the learning rate is 0.005, the batch size is 128, the epoch is 100, and the optimizer is Adaptive Moment Estimation (Adam).

4.3.3. Performance on solving KL divergence vanishing

Fig. 5 describes the learning loss curve of the CVAE and modified CVAE model. In Fig. 5(a), there is an obvious KL vanishing that the penalty loss decreases fast and almost drops to 0 value. Fig. 5(b) describes the curve of the CVAE model after the proposed adaptive loss, where KL penalty loss L_{μ, σ^2} decreases in the early stage and has a trend of KL vanishing. Then, the adaptive weight α decreases, which can reduce the impact of KL loss and make the learner more inclined to the reconstruction error L_{Recon} . By comparison, the reconstruction error of the proposed M-CVAE model is less than the CVAE model. The proposed M-CVAE model can extract more information captured from latent variable z and be beneficial in data augmentation applications.

4.3.4. Visualization on generated samples

We go deeper into the visualization of generated samples. The raw datasets, the CVAE-based data augmentation datasets, and the M-CVAE data augmentation datasets are applied to show the feature visualizations on Task 1 by the t-distributed stochastic neighbor embedding (t-SNE) method. It has a transformation from high-dimensional data to a two-dimension feature space, and then the visualization results are shown in Fig. 6. Based on manifold learning, it relies on unsupervised algorithms to retain the metric knowledge of the original high-dimensional data in dimension reduction. Therefore, the better the visualization effect of t-SNE, the easier the base learner can classify the features.

From a 2-D t-SNE visualization perspective of Fig. 6(a), faulty classes 6 and 7 have the highest number and widest distribution in the original data. Only faulty classes 2 and 3 can have a significant discrepancy from other faulty classes. Besides, faults 4 and 5, with a small sample size, are difficult to distinguish from faulty classes 6 and 7 due to the class imbalance. So, they are easily mistaken for other types of faults in data-driven methods under such the class-imbalance issue.

After CVAE-based data augmentation, the detailed visualization is shown in Fig. 6(b), and many faults can be clearly distinguished. It indicates that generated samples can alleviate the classification difficulties caused by class-imbalance problems. However, some faulty classes 1, 4, 6, and 7 samples still cannot be clearly distinguished. Meanwhile, some generated samples are so far away from the original data (e.g., faulty classes 1 and 2), which can make the learner get misinformation in this case.

M-CVAE alleviates the problem of poor data quality caused by KL vanishing based on adaptive loss improvement, and the visualization is shown in Fig. 6(c). It is clear to distinguish the faulty categories compared with the original and CVAE-augmented data, and in particular, the minority faulty classes 4 and 5 can be clearly distinguished. Therefore, the proposed method can effectively improve the data quality of the original CVAE and enhance the availability of class-imbalance data information.

4.3.5. Visualization on weight calculation

We conduct the KMM method for the weight assignment on generated samples. The generated data far from the original data in the Hilbert space can have a lower weight, close to 0. Data with a closer distribution distance can have higher weights, close to 1. Therefore, for the convenience of visualization, set these data with weights close to 0 as low quality and set data with weights close to 1 as high quality. The two-dimensional visualization of t-SNE corresponding to faulty classes 3 and 5 is shown in Fig. 7. As can be seen from the figure, KMM can assign lower weights to data with lower quality that can affect classification accuracy due to distance, thereby improving the quality of generated data.

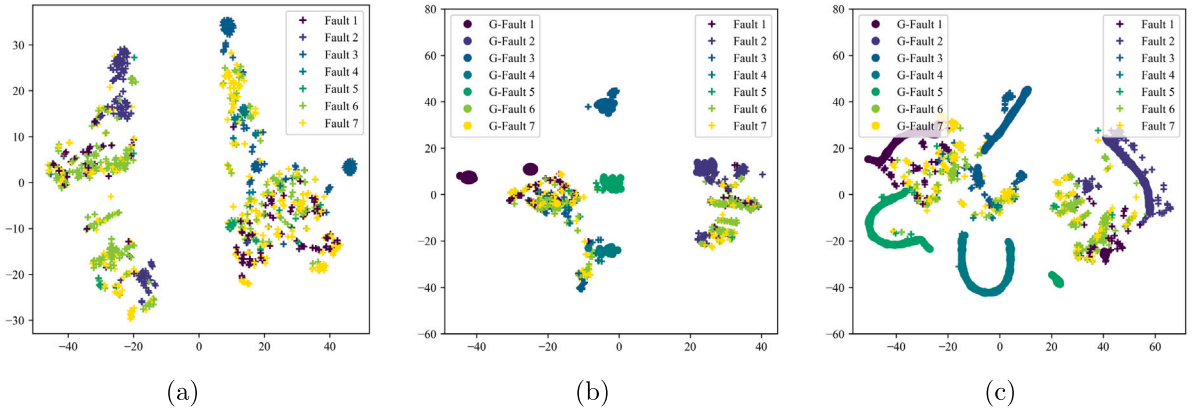


Fig. 6. t-SNE visualization of high-dimensional data from (a) raw datasets, (b) CVAE data augmentation datasets, (c) M-CVAE data augmentation datasets. The ‘plus’ sign represents the original data label (Fault), and the circle represents the generated data label (G-Fault).

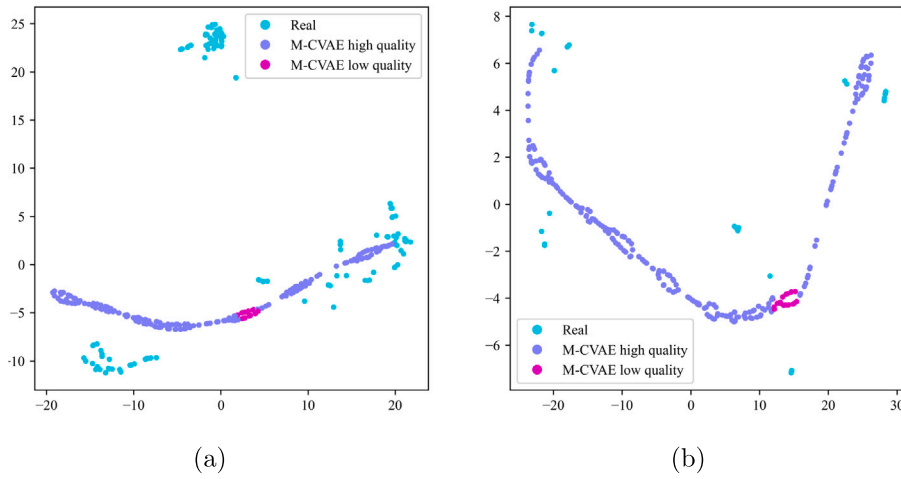


Fig. 7. Visualization of weighted generated data from faulty class 3 (a) and class 5 (b)

4.3.6. Results, analysis, and discussion

Six settings of different imbalance ratios ($Im_r = 10, 20, 30, 50, 60, 100$) are used for comparison. We conduct the WM-CVAE-RF method on class-imbalance fault diagnosis tasks, and the compare results are shown in Table 3. When $Im_r = 10$, the WM-CVAE model gets the best effect, and other data augmentation methods can also improve the performance of the base learner to certain degrees. As the imbalance ratio increases, the advantages of CVAE-based generator methods in dealing with class-imbalance problems are more obvious. Traditional augmented methods show limited improvement in results, and some SOTA methods can yield better results due to the specific designs. In the comparison, deep learning can also show its advantages in results, including the GAN and CVAE models. Based on the KL vanishing consideration, the β -CVAE can also get better results than the CVAE model. Meanwhile, the proposed WM-CVAE data augmentation can get the best results on the average macro-F1-score metric among all experiments and get the best performance in 12 out of 18 metrics.

We summarize the effectiveness for different non-equilibrium ratio datasets in Fig. 8. The increase in the imbalance ratio can lead to a decrease in the performance of the base classifier model. The reason can be attributed to insufficient data information and the fact that the majority of classes influence classification decisions. Besides, the experimental results can verify the robustness of the proposed method under various imbalance ratios.

The confusion matrices in Fig. 9 illustrate the classification results of the WM-CVAE model on tasks 1 and 6. On task 1, the classifier can work well for fault Categories 2, 3, 4, and 6 but less effective for

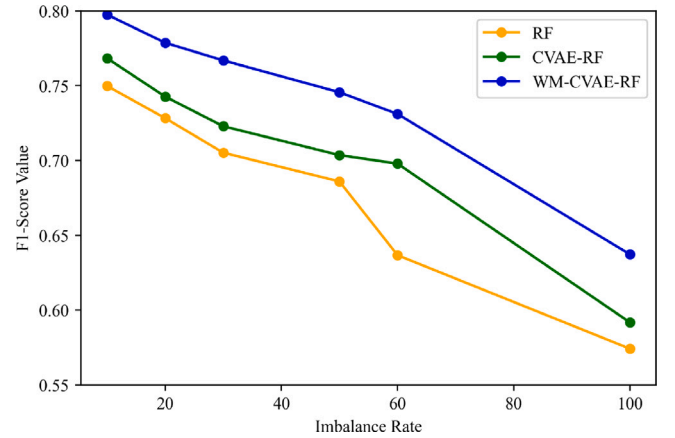


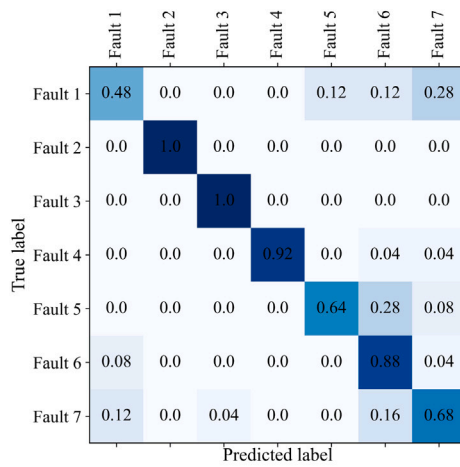
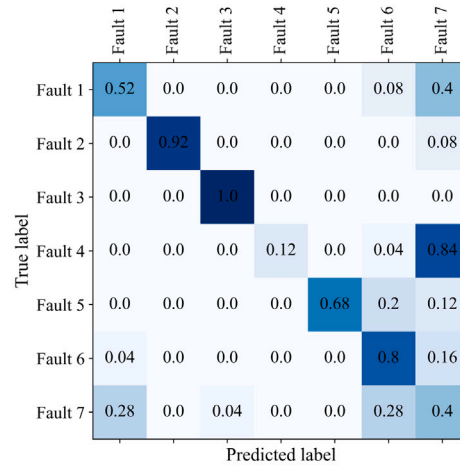
Fig. 8. Experimental results of the data augmentation methods for different imbalance ratio.

Categories 1, 5, and 7. As a minority, some samples of Category 5 are misclassified as Category 6, which indicates that more data information is needed for the base learner. Meanwhile, Categories 1 and 7 have similar metric information and make it difficult for the learner to discriminate between these two types of data, which can be seen in the t-SNE visualization in Fig. 6. On task 6, the number of minority class samples is too small, resulting in the loss of information about

Table 3

Random forest classification results of case study with different data augmentation methods.

Im_r	Evaluator	Data augmentation methods											
		None	ROSE	RUSE	SMOTE	ADASYN	MSDA	EM-LMOT	EF-NCA	GAN	CVAE	β -CVAE	WM-CVAE
10	M_{Re}	0.7486	0.7771	0.7770	0.7828	0.7885	0.7543	0.7886	0.7943	0.7771	0.7657	0.7771	0.8000
	M_{Pr}	0.7851	0.7867	0.7470	0.7957	0.7834	0.7939	0.8123	0.7984	0.8050	0.7907	0.7984	0.8166
	M_{F1}	0.7498	0.7661	0.7503	0.7801	0.7779	0.7514	0.7888	0.7955	0.7776	0.7683	0.7763	0.7975
20	M_{Re}	0.7257	0.6914	0.6171	0.7200	0.7257	0.7371	0.7714	0.7714	0.7543	0.7543	0.7600	0.7829
	M_{Pr}	0.7704	0.7343	0.6753	0.6521	0.7695	0.7828	0.8129	0.8034	0.7918	0.7698	0.7811	0.7971
	M_{F1}	0.7283	0.7017	0.5871	0.6718	0.7294	0.7316	0.7647	0.7740	0.7549	0.7427	0.7578	0.7787
30	M_{Re}	0.7029	0.7371	0.6628	0.7485	0.7257	0.7429	0.7543	0.7600	0.7543	0.7143	0.7429	0.7600
	M_{Pr}	0.7571	0.7756	0.6325	0.7415	0.7342	0.7921	0.7929	0.7975	0.7979	0.7746	0.7701	0.7992
	M_{F1}	0.7052	0.7373	0.6382	0.7332	0.7073	0.7424	0.7540	0.7560	0.7497	0.7229	0.7431	0.7669
50	M_{Re}	0.6857	0.6800	0.6110	0.6857	0.6457	0.6914	0.7200	0.7371	0.7086	0.6971	0.7086	0.7371
	M_{Pr}	0.7659	0.7406	0.5620	0.7319	0.6984	0.7691	0.7825	0.7903	0.7788	0.7677	0.7573	0.7771
	M_{F1}	0.6860	0.6889	0.5618	0.7001	0.6459	0.6922	0.7284	0.7303	0.7164	0.7036	0.7141	0.7456
60	M_{Re}	0.6400	0.6457	0.6342	0.6457	0.6114	0.6857	0.7186	0.7129	0.7086	0.6857	0.6971	0.7200
	M_{Pr}	0.7448	0.7294	0.5943	0.7085	0.6837	0.7672	0.7665	0.7536	0.7774	0.7469	0.7482	0.7609
	M_{F1}	0.6367	0.6527	0.6038	0.6317	0.6052	0.6854	0.7204	0.7278	0.7094	0.6979	0.7077	0.7311
100	M_{Re}	0.5829	0.5828	0.4057	0.6212	0.6114	0.6171	0.6514	0.6471	0.6343	0.6000	0.6343	0.6348
	M_{Pr}	0.7430	0.7198	0.5055	0.7183	0.7374	0.7484	0.7720	0.7595	0.7602	0.7346	0.7625	0.7646
	M_{F1}	0.5742	0.5819	0.3703	0.6172	0.6080	0.5894	0.6245	0.6493	0.6124	0.5918	0.6208	0.6373
Average on M_{F1}		0.6800	0.6881	0.5853	0.6890	0.6790	0.6987	0.7301	0.7388	0.7201	0.7045	0.7200	0.7429

(a) Confusion matrix ($Im_r = 10$)(b) Confusion matrix ($Im_r = 100$)**Fig. 9.** Confusion matrix of the WM-CVAE-RF method for fault classification tasks in Case 1.

Categories 4 and 5 and causing the base learner to misidentify their categories as Category 6 and 7 (majority), which is a crucial factor contributing to its degraded performance on the high imbalance rate.

The WM-CVAE method can be applied to any base classifier, such as support vector machine (SVM), decision tree (DT), eXtreme Gradient Boosting (XGBoost), and multi-layer perceptron (MLP). Therefore, we conduct a comparative study using these base classifiers to verify the superiority and apply RUSE, ROSE, and CVAE methods for comparison. We set the dataset on task 1 with an imbalance rate of 10, and the comparison results are shown in Table 4. In contrast, DT and SVM models perform poorly, MLP model outperform these two models, and XGBoost model perform the best. Under-sampling methods often lead to worse results due to too few minority class samples, and the base classifier cannot acquire enough data from small training samples. The oversampling and CVAE methods can be effectively optimized on the four base classifiers. The WM-CVAE method, which utilizes deep learning, can enhance data quality and achieve superior performance in 10 out of 12 metrics.

We apply the ablation study to determine the improvement of proposed WM-CVAE methods, and the detailed results are shown in Table 5. In the case study, the lack of adaptive loss compared to the

WM-CVAE method can lead to a performance degradation of 3.44% on average. The lack of weight calculation can lead to a performance degradation of 1.57%. Without the proposed optimization methods, only the CVAE-based approach can lead to a performance degradation of 5.16%. And the lack of data augmentation can lead to 8.46% performance drop.

4.4. Case 2: Experiments on multi-phase flow process

In Case 2, we apply the multi-phase flow process for experiments, and the detailed information can be seen in the Ref. [41]. During the industrial process, a particular SCADA system was conducted to collect 24 sensor signals. Based on the real-world dataset, we set two main fault classification tasks corresponding to different imbalance rates with three types of faulty classes and one healthy state. The specific information on experiment settings can be shown in Table 6. On the test set, we set each faulty category to 100 samples. After the same work as Section 4.3.2, we choose a learning rate of 0.005, a batch size of 128, and an epoch of 100 and apply the Adam optimizer.

We validate the effectiveness of the WM-CVAE method, and the experimental results can be shown in Table 7. By comparing experimental results in Case 2, the proposed WM-CVAE method has good

Table 4

Compared experimental results on different basic classifiers.

Basic classifier	Data augmentation	M_{Re}	M_{Pr}	M_{F1}
DT	None	0.7029	0.6910	0.6848
	RUSE	0.6914	0.6611	0.6520
	ROSE	0.7714	0.7177	0.7294
	CVAE	0.7371	0.7428	0.7321
	WM-CVAE	0.7600	0.7627	0.7573
SVM	None	0.6400	0.7288	0.6304
	RUSE	0.6686	0.6823	0.6708
	ROSE	0.6743	0.7359	0.6656
	CVAE	0.7486	0.8094	0.7426
	WM-CVAE	0.7657	0.7787	0.7647
XGBoost	None	0.7714	0.7996	0.7702
	RUSE	0.7657	0.7203	0.7319
	ROSE	0.7829	0.8057	0.7850
	CVAE	0.7886	0.8092	0.7867
	WM-CVAE	0.7886	0.8154	0.7917
MLP	None	0.7257	0.7710	0.7319
	RUSE	0.7029	0.6935	0.6842
	ROSE	0.7600	0.7540	0.7449
	CVAE	0.7771	0.7975	0.7753
	WM-CVAE	0.7829	0.8093	0.7829

Table 5

F1-score results of ablation study in Case 1.

Im_r	None	CVAE	W-CVAE	M-CVAE	WM-CVAE
10	0.7498	0.7683	0.7709	0.7876	0.7975
20	0.7283	0.7427	0.7507	0.7641	0.7787
30	0.7052	0.7229	0.7348	0.7508	0.7669
50	0.6860	0.7036	0.7253	0.7360	0.7456
60	0.6367	0.6979	0.7221	0.7183	0.7311
100	0.5742	0.5918	0.6001	0.6303	0.6373
Average	0.6800	0.7045	0.7173	0.7312	0.7429
Drop	↓ 8.46%	↓ 5.16%	↓ 3.44%	↓ 1.57%	–

Table 6

Description of training set in Case 2.

Class	No. of samples		Description
	Task 5	Task 6	
1	1000	1000	Healthy state
2	50	25	Air line blockage
3	50	25	Water line blockage
4	50	25	Top separator input blockage
Im_r	20	40	

fault classification performance. Meanwhile, from the results of case 2, increasing the class-imbalance rate can significantly reduce the accuracy of the fault classifier because it can be misled to favor the majority class during the training process. In the absence of data information, traditional data enhancement has limited improvement and even adverse effects. Deep learning-based methods can often mine more data information and are not limited to simple augmentation methods, resulting in more accurate classification results. The proposed WM-CVAE model targets low-quality generated pseudo-samples during training, significantly improving the model's accuracy.

4.5. Case 3: Experiments on bearing datasets

We apply the bearing dataset from the University of Cincinnati as a case study [42]. It contains vibration data for four bearings and is sampled with 20,480 points during operation. The faulty conditions can be divided into four categories: health status, inner race, outer race, and ball defects. To determine the health status of bearings, we analyze key features of their vibration signals. These features include the maximum, minimum, mean value, standard deviation, root mean square, skewness, kurtosis, crest factor, and form factor. We set up

Table 7

Experimental results in Case 2 with random forest base classifier.

Task	Im_r	Method	M_{Re}	M_{Pr}	M_{F1}
Task 7	20	None	0.7100	0.5826	0.6289
		RUSE	0.5800	0.5606	0.5690
		ROSE	0.6675	0.5383	0.5901
		CVAE	0.7500	0.6250	0.6667
		β -CVAE	0.7375	0.7798	0.7268
		WM-CVAE	0.8408	0.8132	0.8116
Task 8	40	None	0.5685	0.4930	0.5216
		RUSE	0.6000	0.4719	0.5241
		ROSE	0.6500	0.5223	0.5737
		CVAE	0.6780	0.5520	0.5989
		β -CVAE	0.7475	0.6225	0.6642
		WM-CVAE	0.8020	0.8093	0.7751

Table 8

Experimental results in Case 3 with random forest base classifier.

Task	Im_r	Method	M_{Re}	M_{Pr}	M_{F1}
Task 9	5	None	0.9550	0.9603	0.9548
		RUSE	0.9600	0.9655	0.9597
		ROSE	0.9575	0.9627	0.9572
		CVAE	0.9583	0.9628	0.9575
		β -CVAE	0.9600	0.9646	0.9598
		WM-CVAE	0.9625	0.9666	0.9623
Task 10	10	None	0.9475	0.9545	0.9473
		RUSE	0.9500	0.9573	0.9498
		ROSE	0.9400	0.9474	0.9397
		CVAE	0.9500	0.9563	0.9499
		β -CVAE	0.9500	0.9549	0.9503
		WM-CVAE	0.9575	0.9597	0.9573

a training set with 500 healthy samples. Meanwhile, to show the class imbalance conditions, we set the other three faulty states to 100 samples in Task 9 (resulting in an imbalance rate of 5) and put 50 samples in Task 10 (resulting in an imbalance rate of 10). On the test set, we set each faulty category to 100 samples. We perform the hyper-parameter adjustment experiments in Section 4.3.2 and select the batch size as 128 and the learning rate as 0.005.

The compared experimental results in Case 3 with a random forest classifier are shown in Table 8. The results still show that the imbalance rate significantly affects the training results of the base model. RUSE and ROSE are limited in their ability to improve. Compared to CVAE and β -CVAE, the proposed WM-CVAE can achieve better performance indicators in class-imbalance problems.

4.6. Discussion

The main work is to improve the structure of the CVAE model, which avoids the KL divergence vanishing and reduces the impact of dissimilar generated samples. Unlike the β -CVAE model, an adaptive loss is designed as a 'switch' inside the CVAE structure and automatically changes during different training conditions. When the KL divergence term vanishes, the adaptive weight automatically decreases, and when the KL divergence term rises, the weight increases. It can eliminate the requirement to specify the weight value artificially and allow for the automatic adjustment for different conditions, which is also validated by loss function curves and t-SNE-based data visualization. Besides, we propose the KMM method to assign the higher weights to the high-quality generated samples. Experiments show the performance of the proposed WM-CVAE method in class-imbalance fault classification tasks.

However, the model training and data generation of neural network-based methods are time-consuming, which limits their applicability. Meanwhile, in the case of extreme class imbalance and small sample size, the proposed method misjudges the entire class, as the model can no longer learn essential information from a small amount of data, which can be improved in future works.

5. Conclusion

This paper proposes a WM-CVAE method to solve the class-imbalance fault diagnosis problem, which combines a modified conditional data generator with a KMM-based weighting method. We redesign the parameter updating method in the training process, which can alleviate the existing KL divergence vanishing by an adaptive loss. Then, the KMM method is used to calculate and distribute weights for each generated sample to solve the dissimilarity between the generated and original data. These improvement methods make the data-driven approach more suitable for class-imbalance scenarios, thereby improving the reliability and security of industrial systems. Three real-world case studies are used to verify the performance of the proposed WM-CVAE model. By the loss curve and t-SNE visualization, the WM-CVAE method can solve some disadvantages of standard CVAE-based methods, including the KL divergence vanishing issue and the negative effect of low-quality generated samples. Diagnostic results show that the proposed method improves the performance of base learners under higher imbalance ratio conditions. In the future, small & imbalanced data intelligent fault diagnosis (S&I-IFD) is our direction, as it is common in practical industrial applications.

CRedit authorship contribution statement

Jilun Tian: Writing – original draft, Visualization, Validation, Methodology, Conceptualization. **Yuchen Jiang:** Writing – review & editing, Funding acquisition. **Jiushi Zhang:** Methodology, Conceptualization. **Hao Luo:** Writing – review & editing, Supervision, Funding acquisition. **Shen Yin:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data used in the research is public dataset.

Acknowledgments

This work was supported in part by the Natural Science Foundation of Heilongjiang Province of China under Grant YQ2022F009, in part by the National Natural Science Foundation of China (NSFC) under Grant 62203143, and in part by the Heilongjiang Province Postdoctoral Foundation under Grant LBH-Z22130.

References

- [1] Zio E. Prognostics and Health Management (PHM): Where are we and where do we (need to) go in theory and practice. *Reliab Eng Syst Saf* 2022;218:108119.
- [2] Wu J, Zhao Z, Sun C, Yan R, Chen X. Learning from class-imbalanced data with a model-agnostic framework for machine intelligent diagnosis. *Reliab Eng Syst Saf* 2021;216:107934.
- [3] Leoni L, De Carlo F, Abaei MM, BahooToroody A, Tucci M. Failure diagnosis of a compressor subjected to surge events: A data-driven framework. *Reliab Eng Syst Saf* 2023;233:109107.
- [4] Chaleshtori AE, Aghaie A. A novel bearing fault diagnosis approach using the Gaussian mixture model and the weighted principal component analysis. *Reliab Eng Syst Saf* 2024;242:109720.
- [5] de Andrade Melani AH, de Carvalho Michalski MA, da Silva RF, de Souza GFM. A framework to automate fault detection and diagnosis based on moving window principal component analysis and Bayesian network. *Reliab Eng Syst Saf* 2021;215:107837.
- [6] Santos P, Maudes J, Bustillo A. Identifying maximum imbalance in datasets for fault diagnosis of gearboxes. *J Intell Manuf* 2018;29:333–51.
- [7] Jiang Y, Yin S, Kaynak O. Performance supervised plant-wide process monitoring in industry 4.0: A roadmap. *IEEE Open J Ind Electron Soc* 2021;2:21–35.
- [8] Miao J, Wang J, Miao Q. An enhanced multifeature fusion method for rotating component fault diagnosis in different working conditions. *IEEE Trans Reliab* 2021;70(4):1611–20.
- [9] Jiang Y, Wu S, Yang H, Luo H, Chen Z, Yin S, Kaynak O. Secure data transmission and trustworthiness judgement approaches against cyber-physical attacks in an integrated data-driven framework. *IEEE Trans Syst Man Cybern: Syst* 2022;52(12):7799–809.
- [10] Tian J, Jiang Y, Zhang J, Wu S, Luo H. A novel transfer ensemble learning framework for remaining useful life prediction under multiple working conditions. *IEEE Trans Instrum Meas* 2023;1.
- [11] Zhang J, Tian J, Alcaide AM, Leon JI, Vazquez S, Franquelo LG, Luo H, Yin S. Lifetime extension approach based on the Levenberg–Marquardt neural network and power routing of DC–DC converters. *IEEE Trans Power Electron* 2023;38(8):10280–91.
- [12] Shi P, Wu S, Xu X, Zhang B, Liang P, Qiao Z. TSN: A novel intelligent fault diagnosis method for bearing with small samples under variable working conditions. *Reliab Eng Syst Saf* 2023;109575.
- [13] Li W, Huang R, Li J, Liao Y, Chen Z, He G, Yan R, Gryllias K. A perspective survey on deep transfer learning for fault diagnosis in industrial scenarios: Theories, applications and challenges. *Mech Syst Signal Process* 2022;167:108487.
- [14] He H, Garcia EA. Learning from imbalanced data. *IEEE Trans Knowl Data Eng* 2009;21(9):1263–84.
- [15] Ding Y, Jia M, Zhuang J, Cao Y, Zhao X, Lee C-G. Deep imbalanced domain adaptation for transfer learning fault diagnosis of bearings under multiple working conditions. *Reliab Eng Syst Saf* 2023;230:108890.
- [16] Buda M, Maki A, Mazurowski MA. A systematic study of the class imbalance problem in convolutional neural networks. *Neural Netw* 2018;106:249–59.
- [17] Qian M, Li Y-F. A novel adaptive undersampling framework for class-imbalance fault detection. *IEEE Trans Reliab* 2022;(99):1–15.
- [18] Sharma S, Bellinger C, Krawczyk B, Zaiane O, Japkowicz N. Synthetic oversampling with the majority class: A new perspective on handling extreme imbalance. In: 2018 IEEE international conference on data mining (ICDM). IEEE; 2018, p. 447–56.
- [19] Jang K, Hong S, Kim M, Na J, Moon I. Adversarial autoencoder based feature learning for fault detection in industrial processes. *IEEE Trans Ind Inf* 2021;18(2):827–34.
- [20] Zemouri R, Ibrahim R, Tahan A. Hydrogenerator early fault detection: Sparse Dictionary Learning jointly with the Variational Autoencoder. *Eng Appl Artif Intell* 2023;120:105859.
- [21] Chai Z, Zhao C, Huang B, Chen H. A deep probabilistic transfer learning framework for soft sensor modeling with missing data. *IEEE Trans Neural Netw Learn Syst* 2021;33(12):7598–609.
- [22] Yi H, Jiang Q, Yan X, Wang B. Imbalanced classification based on minority clustering synthetic minority oversampling technique with wind turbine fault detection application. *IEEE Trans Ind Inf* 2020;17(9):5867–75.
- [23] Zhang T, Chen J, Li F, Zhang K, Lv H, He S, Xu E. Intelligent fault diagnosis of machines with small & imbalanced data: A state-of-the-art review and possible extensions. *ISA Trans* 2022;119:152–71.
- [24] Li B, Tang B, Deng L, Wei J. Joint attention feature transfer network for gearbox fault diagnosis with imbalanced data. *Mech Syst Signal Process* 2022;176:109146.
- [25] Khan SH, Hayat M, Bennamoun M, Sohail FA, Togneri R. Cost-sensitive learning of deep feature representations from imbalanced data. *IEEE Trans Neural Netw Learn Syst* 2017;29(8):3573–87.
- [26] Wu Z, Zhang H, Guo J, Ji Y, Pecht M. Imbalanced bearing fault diagnosis under variant working conditions using cost-sensitive deep domain adaptation network. *Expert Syst Appl* 2022;193:116459.
- [27] Qian M, Li Y-F. A weakly supervised learning-based oversampling framework for class-imbalanced fault diagnosis. *IEEE Trans Reliab* 2022;71(1):429–42.
- [28] San Martin G, Lopez Droguett E, Meruane V, das Chagas Moura M. Deep variational auto-encoders: A promising tool for dimensionality reduction and ball bearing elements fault diagnosis. *Struct Health Monit* 2019;18(4):1092–128.
- [29] Pol AA, Berger V, Germain C, Cerminara G, Pierini M. Anomaly detection with conditional variational autoencoders. In: 2019 18th IEEE international conference on machine learning and applications (ICMLA). IEEE; 2019, p. 1651–7.
- [30] Khanam S, Ahmedy I, Idris MYI, Jaward MH. Towards an effective intrusion detection model using focal loss variational autoencoder for internet of things (IoT). *Sensors* 2022;22(15):5822.
- [31] Zhao X, Yao J, Deng W, Jia M, Liu Z. Normalized conditional variational auto-encoder with adaptive focal loss for imbalanced fault diagnosis of bearing-rotor system. *Mech Syst Signal Process* 2022;170:108826.
- [32] Tolstikhin I, Bousquet O, Gelly S, Schoelkopf B. Wasserstein auto-encoders. 2017, arXiv preprint arXiv:1711.01558.
- [33] Wang C, Xin C, Xu Z, Qin M, He M. Mix-VAEs: A novel multisensor information fusion model for intelligent fault diagnosis. *Neurocomputing* 2022;492:234–44.
- [34] Zhu Q, Su J, Bi W, Liu X, Ma X, Li X, Wu D. A batch normalized inference network keeps the kl vanishing away. 2020, arXiv preprint arXiv:2004.12585.
- [35] Fu H, Li C, Liu X, Gao J, Celikyilmaz A, Carin L. Cyclical annealing schedule: A simple approach to mitigating kl vanishing. 2019, arXiv preprint arXiv:1903.10145.

- [36] Huang J, Gretton A, Borgwardt K, Schölkopf B, Smola A. Correcting sample selection bias by unlabeled data. *Adv Neural Inf Process Syst* 2006;19.
- [37] Kingma DP, Welling M. Auto-encoding variational bayes. 2013, arXiv preprint arXiv:1312.6114.
- [38] Sohn K, Lee H, Yan X. Learning structured output representation using deep conditional generative models. *Adv Neural Inf Process Syst* 2015;28.
- [39] Xiao Y, Liang F, Liu B. A transfer learning-based multi-instance learning method with weak labels. *IEEE Trans Cybern* 2020;52(1):287–300.
- [40] Asuncion A, Newman D. UCI machine learning repository. 2007.
- [41] Ruiz-Cárcel C, Cao Y, Mba D, Lao L, Samuel R. Statistical process monitoring of a multiphase flow facility. *Control Eng Pract* 2015;42:74–88.
- [42] Brito LC, Susto GA, Brito JN, Duarte MAV. Fault Diagnosis using eXplainable AI: A transfer learning-based approach for rotating machinery exploiting augmented synthetic data. *Expert Syst Appl* 2023;120860.
- [43] Lemaître G, Nogueira F, Aridas CK. Imbalanced-learn: A python toolbox to tackle the curse of imbalanced datasets in machine learning. *J Mach Learn Res* 2017;18(1):559–63.
- [44] He H, Bai Y, Garcia EA, Li S. ADASYN: Adaptive synthetic sampling approach for imbalanced learning. In: 2008 IEEE international joint conference on neural networks (IEEE world congress on computational intelligence). IEEE; 2008, p. 1322–8.
- [45] Wu Z, Lin W, Fu B, Guo J, Ji Y, Pecht M. A local adaptive minority selection and oversampling method for class-imbalanced fault diagnostics in industrial systems. *IEEE Trans Reliab* 2019;69(4):1195–206.
- [46] Sun Q, Peng F, Yu X, Li H. Data augmentation strategy for power inverter fault diagnosis based on wasserstein distance and auxiliary classification generative adversarial network. *Reliab Eng Syst Saf* 2023;237:109360.
- [47] Dewangan G, Maurya S. Fault diagnosis of machines using deep convolutional beta-variational autoencoder. *IEEE Trans Artif Intell* 2021;3(2):287–96.